

Assessment of Recyclable thermoplastic material for HVDC cable application

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Abstract— One of the major challenges in power transmission is the losses incurred during transmitting power from the isolated power generation source to substation. Transmission through High Voltage Direct Current (HVDC) is an economically reliable alternative and a promising solution to minimize losses over long distance transmission. The aim of our work is to replace the conventionally used cross linked poly ethylene (thermo set material) by an alternative thermoplastic material in order to increase the life of the cable. Thermoplastic materials are highly desirable for HVDC cable insulation because of its recyclability and absence of by products. In this work the dielectric properties of the isotactic polypropylene (iPP) is improved by introduction of inorganic nanoparticles. Investigation is done on the break down strength of the iPP/nano composites thin film with different weight percentage for HVAC and HVDC applications. A comprehensive analysis of iPP/nano composites its suitability in HVDC cable insulation application.

Keywords—polypropylene, nanocomposite, insulation material, dielectric, breakdown strength, thermoplastic;

I. INTRODUCTION

The power demand in India is growing day by day and the impact of CO₂ emission in the environment makes a drastic climatic change [1, 2 and 3]. There is a need of an alternative such as renewable source to compensate the growing demand. The growing demand is expected to be met by the renewable energy and the whole world is actively working on it. Many power generation (wind energy, solar energy) units are located far from the centre of demand. It is challenging to transmit the power from generating unit to consumer end without losses. HVDC cable transmission has overcome the demerits of the HVAC system. The first HVDC cable was put into operation in 1954, connecting Gotland Island and Swedish main land. Due to the emergence of renewable energy sources, HVDC transmission is the choice as the bulk power needs to be transported from off shore to the converter end at reduced loss [4, 5 and 6].

Insulation plays a major role in determining the life of the underground cable. At early stage mass impregnated and oil immersed paper was used as the insulation. Later LDPE and HDPE were used as the insulation material for HVDC cable insulation. Due to poor mechanical properties and low with stand temperature LDPE is replaced by the cross linking byproducts. XLPE was the most widely used underground cable insulation, the first application of XLPE DC cable was the Gotland project, Sweden operating at 80kV in 1998. Higher operating voltage of 500kV HVDC extruded cable is developed in Japan but not yet commercially used. XLPE

byproducts could induce severe space charge accumulation which will degrade the life of the HVDC cable. XLPE is a thermoset material which is difficult to recycle and the byproducts such as phenol, cumyl alcohol, acetophenone and methyl styrene introduced during cross linking would degrade the electrical properties and life time of the cable insulation [7, 8]. As protection of global environment is the main concern. Cross linked polyethylene are suited as they are not favorable to reduce, recycle and reuse. The choice of selecting, Isotactic polypropylene is because it has a high melting point and also enhanced electrical properties such as breakdown strength and dc volume resistivity. It is a thermoplastic material, so it can be easily recyclable [9].

Nanoparticles have many special effects such as surface effects, quantum effects and small size effects. It has been widely used to improve the breakdown strength and space charge properties of the HVDC cable insulation. Nanometer size particle exhibits novel properties according to length and surface area compared to the bulk material. Nanocomposites are known to improve the electrical performance of the insulation material if optimum concentration is added. Most of the researches are towards the modification of dielectric properties of LDP and XLPE with nano particles but only few articles deal with thermoplastic materials [10]. Polymer based nanocomposites draws more interest in recent times. The use of polymer is well known, however more research is towards the polymer nanocomposites, to develop improved insulating materials.

In this work, we did a study on iPP with three different nanoparticles such as TiO₂, MgO and ZnO. Investigation is mainly on effects nanoparticles whether it enhances the electrical properties of the insulation material. From the test data, the breakdown strength of the iPP with nano filler is more promising for the thermoplastic HVDC cable insulation.

II. EXPERIMENTAL SETUP

A. Sample preparation

The iPP is used as base material which is in pellet form. The iPP pellets are changed into the liquid phase using xylene as a solvent at temperature of elevated temperature. Three different Nanoparticles (average diameter of 50nm) with various weight percentages (1, 2, 3, 4 and 5wt %) are mixed to the base material in the magnetic stirrer at 1500rpm to get homogeneous form. A neat thin film and nanocomposite thin

film of thickness 100μm is prepared using of spin coating technique. Thin films are subjected to both AC and DC stress.

B. Measurements

The morphology of the thin films and nanocomposites are measured by the Scanning Electron Microscope (SEM) model S-3400N (Hitachi make).

Break down test was carried out on the thin film sample at room temperature. The electric field was applied until the break down failure, five specimens were tested for each nanocomposites and the average value is taken as the break down strength.

C. Experimental setup

The test sample was inserted between the sphere-plane electrodes. The upper sphere-electrode diameter of 20mm is connected to the high voltage and the bottom plane electrode area of 70mmX30mm with thickness 15mm is connected to the ground.

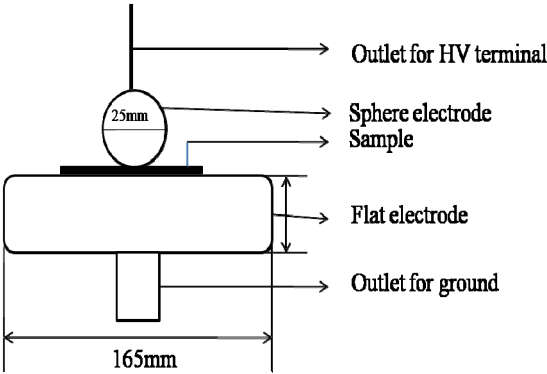


Figure 1. Electrode system of the test setup

The specification of the sphere plane electrode is shown in the Figure 1 as per IEC 60243-I standard.

The output from the high voltage transformer (10KVA, 230V/11kV, 50Hz) is given to the sphere electrode and the bottom flat electrode is grounded. The breakdown strength of the insulation is determined by applying the voltage until the test specimens gets break down.

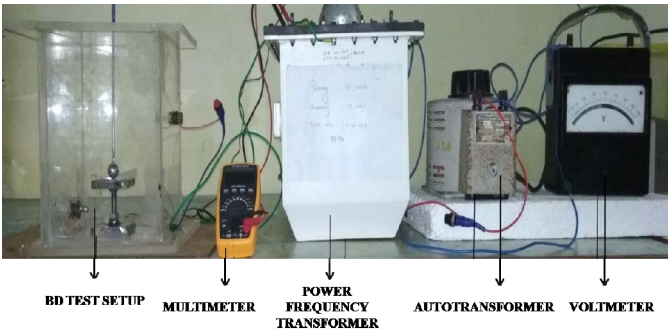


Figure 2. Experimental setup of HVAC system

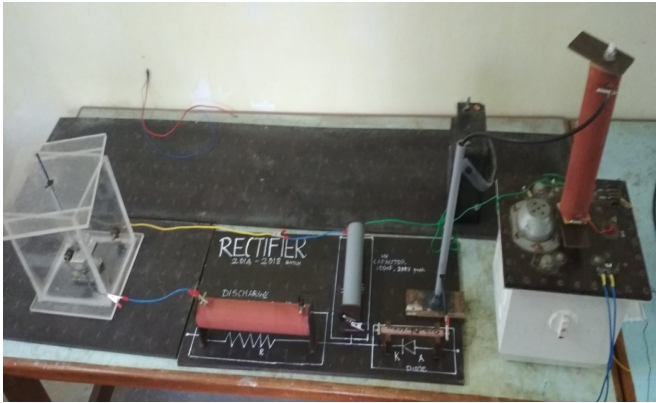
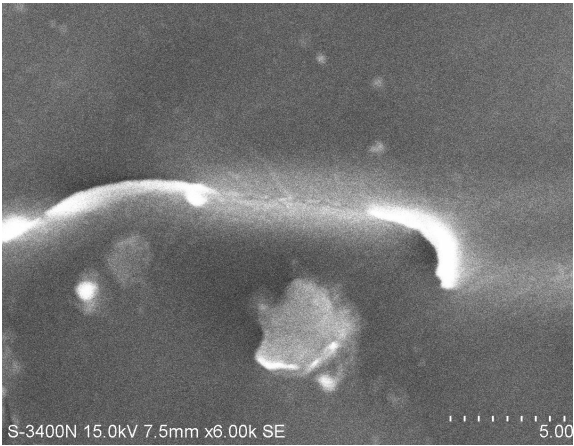


Figure 3. Experimental setup of HVDC system

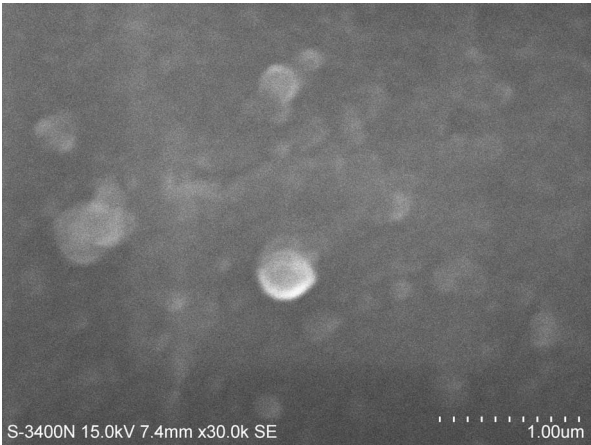
This test is carried at room temperature. Figure 2 and 3 shows the experimental setup of AC and DC breakdown test. For the HVDC the rectified output from the circuit is stressed the sample.

III. RESULTS AND DISCUSSION

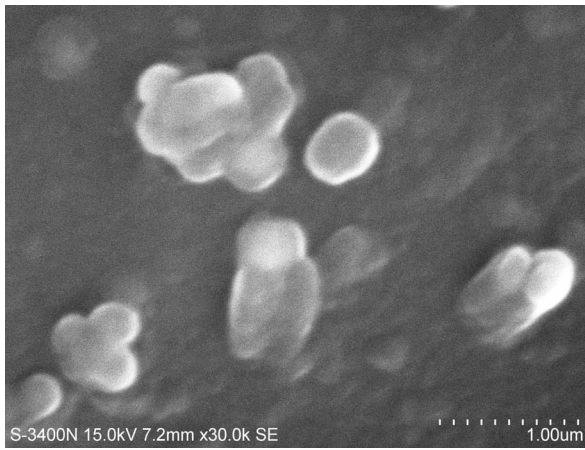
The dispersion of nanoparticles in the iPP matrix is characterized by scanning electron microscope.



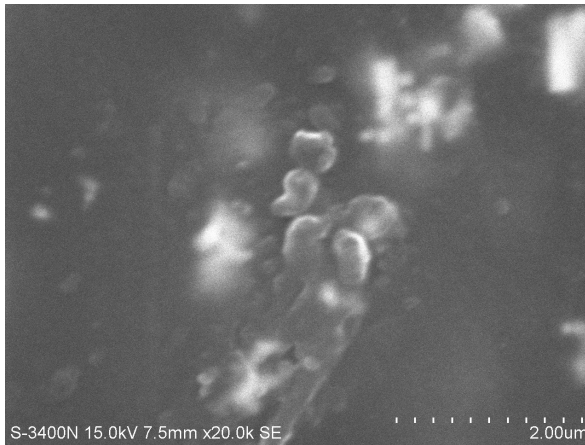
(a)



(b)



(c)



(d)

Figure 4. The SEM image of (a)iPP (b)TiO₂(c) MgO and (d) ZnO composites

The nanoparticles are dispersed in the sample however the size is in the range of 170nm, 230nm, 210nm due to the nature of particles and may be due to insufficient dispersion. The SEM image of iPP and nanocomposite film is shown in Figure 4.

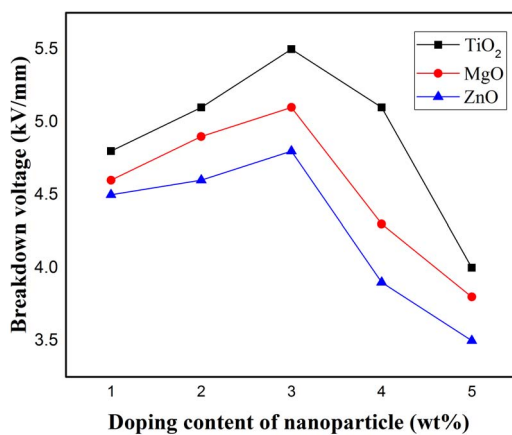


Figure 5. Comparative analysis of HVAC breakdown strength of different nanoparticles

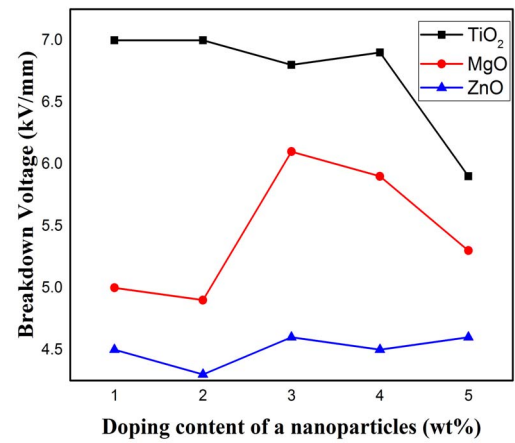


Figure 6. Comparative analysis of HVDC breaks down strength of different nanoparticles

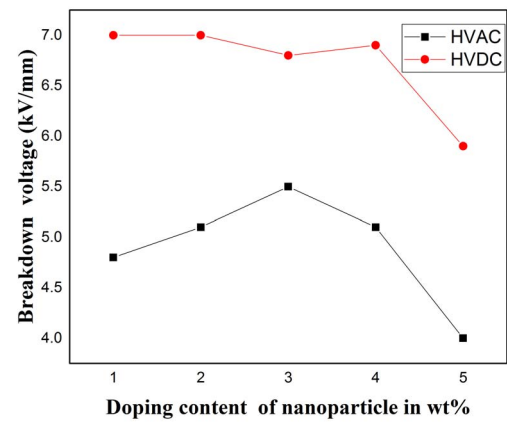


Figure 7. Comparative analyses of TiO₂ breakdown characteristics in HVAC and HVDC

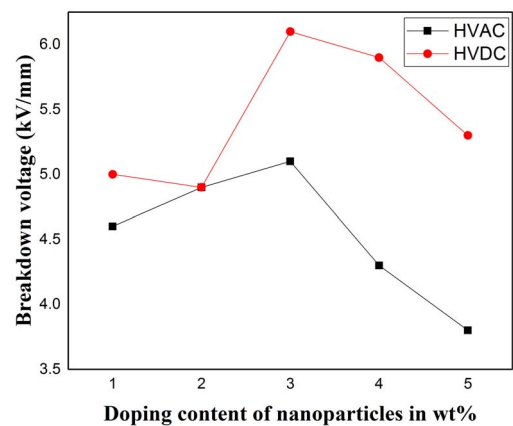


Figure 8. Comparative analyses of MgO breakdown characteristics in HVAC and HVDC

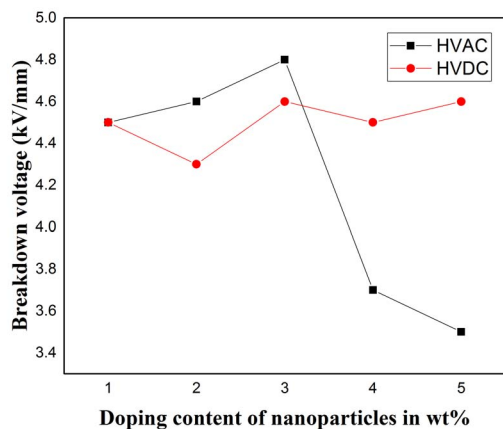


Figure 9. Comparative analyses of ZnO breakdown characteristics in HVAC and HVDC

Break down strength of the iPP film and nanocomposites are shown in Figure 5 to 9. From the data the Figure 5 shows that, the break down strength increases with increase in nanoparticle up to 3% and gets decreased if concentration is increased.

The breakdown characteristics of the nanocomposites under DC stress is shown in the Figure 6, as the concentration is increased breakdown strength of TiO₂ almost decreased gradually up to 4% and decreased abruptly from 4 to 5%. In MgO Nanocomposite initially decreased and then a drastic change from 2% to 3% and then start to decrease. The doping of TiO₂ and MgO, show that most desirable space charge suppression up to 3phr [10].

Breakdown characteristics of ZnO nanocomposites initially decreased from 1% to 2% and then increased to some extent. In the small concentration the ZnO nanoparticles observed the injected charges near the cathode and it prevents the extraction of electron, it is promising that suppression of space charge at 5% [11].

At higher concentration the nanoparticles get agglomeration and the inter-particles distance decreases which will create a low resistance path. So the Polymer matrix will reduce its breakdown strength of the sample. Among these three kinds of nanoparticles, TiO₂ nanoparticles shows higher break down strength than MgO and ZnO. From the Fig 7, 8 and 9, the TiO₂ shows excellent result comparative to other

nanoparticles and also it exhibits improved electrical characteristics to neat polymer material. HVDC exhibits higher breakdown strength over HVAC.

IV. CONCLUSION

From the test data, it is evident that the break down strength of the thermoplastic material is elevated by adding the nano particles. TiO₂ nano particles enhance the electrical property than the MgO and ZnO.

From the comparative analysis the break down strength of the HVDC is higher than that of the HVAC. It is clear evident that the iPP/TiO₂ nanocomposite is promising material for HVDC cable insulation system.

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